HANDSHAKE

PROJECT NAME:	Receiver and Transmitter using NRF24L Module and STM Microcontroller
PROJECT GROUP:	Sahil Khandare & Pratik Khadse

WHY THIS PROJECT ?

- This project was undertaken to create an alternative to the existing Flysky transmitter– receiver system. Our goal is to design and develop a custom transmitter and receiver tailored specifically to our requirements. These requirements will evolve based on the future competitions the club participates in.
- Our mission is to develop and build all electronic components in-house, ensuring full control over performance, customization and innovation.

INTRODUCTION

- This project is developed using an STM microcontroller paired with the NRF24L01 wireless module. Both the PCB and the firmware were designed and built entirely in-house. The NRF24L01 was selected primarily for its high data rate, which supports fast and efficient communication between the transmitter and receiver. However, during development, we realized that this choice came with a trade-off—our focus on speed led us to overlook the module's limited operational range.

KEY FEATURES

1. Custom Firmware Development:
   Entire firmware written in-house, allowing complete control over optimization.
2. In-House PCB Development:
   Custom-designed PCB tailored specifically for the transmitter–receiver architecture.
3. High Data Rate Communication:
   Utilizes the NRF24L01 module's fast data-rate capability for efficient and low-latency communication.

ARCHITECTURE

1. Microcontroller Integration:
   Rx - STM32G070RBT6
   Tx - STM32G030KBT6
2. Wireless Module Selection:
   NRF24L01
3. PCB Material :
   Copper Clad using Acid Etching.

RESULTS

- The custom transmitter–receiver system successfully established stable communication using the STM microcontroller and NRF24L01 module, with the high data rate delivering fast and responsive performance.
- The in-house PCB fabricated through acid-wash etching on copper-clad boards operated reliably, supporting clean signal routing and consistent power delivery. However, testing revealed a notable limitation in communication range, as the emphasis on high data rate reduced overall distance performance.

FUTURE SCOPE

1. ☐ **Use of HC-12**
2. ☐ **Multi Rx with Single Tx**
3. ☑ **Going for OSHWA certification**